

USE CASE analysis

Use Case : Charging Slot

Primary Actor: EV Driver

Supporting Actor: Notification Service

Preconditions

1. The EV Driver has access to the EV Charging Station Reservation System.
2. At least one charging station has an available charging socket and reservable time slot.
3. The driver's vehicle connector type is available or provided to the system for compatibility checking.
4. The selected socket and time slot have not already been reserved or blocked.

Postconditions

5. A compatible charging socket and time slot are reserved for the EV Driver.
6. The reservation is stored with the selected station, socket, date, time, and applicable dynamic power rate.
7. The reserved socket and time slot are no longer available for conflicting reservations.
8. A reservation confirmation is sent to the EV Driver.

Main Success Scenario

9. The EV Driver opens the charging station reservation page.
10. The system displays available charging stations and charging sockets.
11. The EV Driver provides or selects the vehicle connector type.
12. The system identifies and displays compatible available charging sockets.
13. The EV Driver selects a compatible charging socket.
14. The system displays the available charging time slots and applicable dynamic power rates.
15. The EV Driver selects a preferred available time slot.
16. The system verifies that the selected socket is compatible and that the selected slot is still available.
17. The EV Driver confirms the reservation.
18. The system creates and saves the charging reservation.
19. The system marks the selected socket and time slot as unavailable for conflicting reservations.
20. The system includes the Send Reservation Confirmation use case.
21. The Notification Service sends the EV Driver a confirmation containing the station, socket, date, time, and reservation details.
22. The system displays a successful reservation confirmation to the EV Driver.
23. The use case ends successfully.

Alternate Flow — Selected Socket or Slot Is No Longer Available

24. The system detects that the selected socket is unavailable or the selected time slot has already been reserved or blocked.
25. The system informs the EV Driver that the selected charging option is no longer available.
26. The system displays the currently available compatible sockets and time slots.
27. The EV Driver selects another available compatible socket or time slot.
28. The flow returns to Step 8 of the Main Success Scenario.
29. If no suitable option is selected, the reservation process is cancelled and no reservation is created.